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1 Methodology

Since our objective is to study family size, there is no direct variable in the dataset that represents this concept. Therefore, we assume that each household in the sample has the same number of elder family members, including parents and grandparents. This assumption allows us to use "Number of Children" as a proxy for family size and treat it as our response variable.Since "Number of Children" is clearly a positive integer, we first considered using Poisson regression as our modeling approach. However, since each observation has a different exposure time, we need to ensure the fairness of the model by making the count data from different observational units comparable. To achieve this, we introduce an offset variable to represent the exposure level. Specifically, we divide "Number of Children" by this offset, transforming our response variable into a ratio-based measure, ensuring that the model reasonably explains the event occurrence under a per-unit exposure framework.

Next, we validated the assumptions of Poisson regression to ensure the appropriateness of our model. First, we checked whether the log-mean of the response variable exhibited a linear relationship with the predictor variables by examining the QQ-plot to see if the points closely followed the reference line. Next, we verified the independence of observations. To assess equidispersion, we calculated the dispersion parameter: $\hat{\phi} = \frac{\sum(\text{Pearson residuals})^2}{\text{Degrees of Freedom for Pearson Residuals}}$. If $\hat{\phi}$ significantly exceeded 1, we concluded that overdispersion was present in the data and considered alternative models, such as negative binomial regression. Finally, we confirmed that our response variable consisted of non-negative integer counts, ensuring its suitability for Poisson regression.

Upon validating the model's applicability, we conducted a hypothesis test to evaluate the statistical significance of the predictor variables and assess the robustness of the model's predictions. The hypotheses are formulated as follows:

$$H_0 : \beta_i = 0, \quad H_1 : \beta_i \neq 0$$

where β_i denotes the regression coefficient of the predictor variable under examination.

The statistical inference is based on the z-statistic, computed as:

$$z = \frac{\hat{\beta}_i}{\text{SE}(\hat{\beta}_i)} \sim N(0, 1)$$

where $\hat{\beta}_i$ represents the maximum likelihood estimate of β_i, and $\text{SE}(\hat{\beta}_i)$ is the standard error of the estimated coefficient.

To quantify the uncertainty associated with the parameter estimates, the 95% confidence interval (CI) for the exponentiated coefficient, which corresponds to the rate ratio in Poisson regression, is given by:

$$\exp(\text{CI}_{95\%}) = \left[\exp\left(\hat{\beta}_i - 1.96 \times \text{SE}(\hat{\beta}_i)\right), \exp\left(\hat{\beta}_i + 1.96 \times \text{SE}(\hat{\beta}_i)\right)\right]$$

If the confidence interval excludes 1, it indicates that the predictor variable exerts a statistically significant influence on the response variable at the 95% confidence level. This suggests that changes in the predictor are associated with changes in the expected count of the response variable, supporting its relevance in the model.

2 Result

Table 1 presents the statistical summary of all important variables. It is shown that the distribution of The Number of Children is right-skewed, with a median and mean of 2. The interquartile range (1 to 3) shows that half of the population falls within this range, while the maximum value of 17 suggests high variability. The majority of individuals marry between the ages of 18 and 25, with the highest concentration in the 22–25 age group (1,468 individuals), followed by those aged 20–22 (1,126 individuals). Regarding the duration of marriage, the range extends from 0 months to 431 months. The median marriage duration is 148 months, with a mean of 155.2 months. However, the slightly higher mean contributes to a mild right-skewed distribution. In terms of literacy, the vast majority of the population is literate (4,567 individuals), whereas only 581 individuals are illiterate.

Table 1: Summary of Key Variables

Number of Children	Months Since Marriage	Count by Age at Marriage	Count by Literate or Illiterate
Min: 0	Min: 0.0	At 0-15: 52	Literate: 4567
1st Qu: 1	1st Qu: 73.0	At 15-18: 452	Illiterate: 581
Median: 2	Median: 148.0	At 18-20: 910	
Mean: 2	Mean: 155.2	At 20-22: 1126	

Table 1: Summary of Key Variables

Number of Children	Months Since Marriage	Count by Age at Marriage	Count by Literate or Illiterate
3rd Qu: 3 Max: 17	3rd Qu: 230.0 Max: 431.0	At 22-25: 1468 At 25-30: 923 At over 30: 217	

Recall our research objective: To explore the impact of literacy and age at marriage on family size (i.e., the number of children). We initially considered using Poisson regression because the number of children, as the response variable, is a non-negative integer. We included "Age at Marriage" and "Literate or Illiterate" as our predictor variables. Additionally, in (**xxxx?**)'s study, we have already found that the duration of marriage has a significant effect on the number of children. To control for the influence of different marriage durations on fertility, ensuring that the model reflects the independent effects of literacy and age at marriage, we included "Months of Marriage" as an offset to standardize the exposure period (i.e., the duration of marriage) across observations.

It is important to note that, since the link function of the Poisson distribution is log, directly taking log(0) when constructing the offset would result in negative infinity, which could affect the model's stability and accuracy. Therefore, when constructing the offset, we ensured that the number of months of marriage for each observation was at least 1 (even if the original data contained 0s, this adjustment prevents the log(0) issue). Additionally, we divided the months by 12 to convert the time scale to years, making the model's time representation more reasonable. By incorporating the offset, the original response variable (the number of children) is essentially transformed into a rate-based metric in the model. This means that we are now estimating the fertility rate per unit of time (e.g., per year) rather than the total number of children. With the logarithmic link function, this rate is expressed in logarithmic form, allowing the model's results to be interpreted as the logarithm of the average number of children per year after marriage.

Furthermore, based on Table 1, we observed that the majority of our sample got married between the ages of 22 and 25. Therefore, we set the base level for the "Age at Marriage" variable to the "22 to 25 years" group. This decision allows us to interpret the effects of other age-at-marriage categories relative to the most common marriage age group in our sample.

Thus, our model is:

$$
\begin{aligned}
\log(E[\text{Children}_i]) = &\beta_0 + \beta_1 \text{M}_{0to15}(i) + \beta_2 \text{M}_{15to18}(i) + \\
&\beta_3 \text{M}_{18to20}(i) + \beta_4 \text{M}_{20to22}(i) + \beta_5 \text{M}_{25to30}(i) + \\
&\beta_6 \text{M}_{30toInf}(i) + \beta_8 \text{Literacy}_i + \\
&\log(\text{Months}_i) + \epsilon_i
\end{aligned}
$$

Where:

$$
\begin{aligned}
\text{Children}_i &: \text{ The total number of children born to the } i\text{th individual.} \\
M_{\text{ato}b}(i) &: \text{ Binary indicator variable, taking the value of 1 if the } i\text{th individual} \\
&\quad \text{entered into marriage between ages } a \text{ and } b, \text{ and 0 otherwise.} \\
\text{Literacy}_i &: \text{ Binary indicator variable, equal to 1 if the } i\text{th individual is illiterate,} \\
&\quad \text{and 0 otherwise.} \\
\text{Months}_i &: \text{ The total duration of marriage (in months) for the } i\text{th individual.} \\
\beta_0, \beta_1, \ldots, \beta_8 &: \text{ Coefficients estimated in the regression model.} \\
\epsilon_i &: \text{ Stochastic error term capturing unobserved heterogeneity and random fluctuations.}
\end{aligned}
$$

After determining the initial model, we examined its assumptions. We first observed the QQ plot (Figure 1) and found that most of the residuals closely follow the reference line, indicating that the deviance residuals are approximately normally distributed. This suggests that the Poisson model is a reasonable fit for the data and confirms that the model satisfies the assumption that the log-mean of the response variable is linearly related to the predictor variables. Then, we calculated the dispersion parameter $\hat{\phi}$:

$$
\hat{\phi} = \frac{\sum(\text{Pearson residuals})^2}{\text{Degrees of Freedom for Pearson Residuals}} = \frac{5600.133}{5140} = 1.08952 \approx 1
$$

$\hat{\phi} \approx 1$ indicates that the covariates in our model explain most of the variation, which suggests that the data do not exhibit significant overdispersion, and thus there is no need to switch the model to a negative binomial. Additionally, the assumptions that the observations are independent and that the response variable consists of positive integers are straightforward.

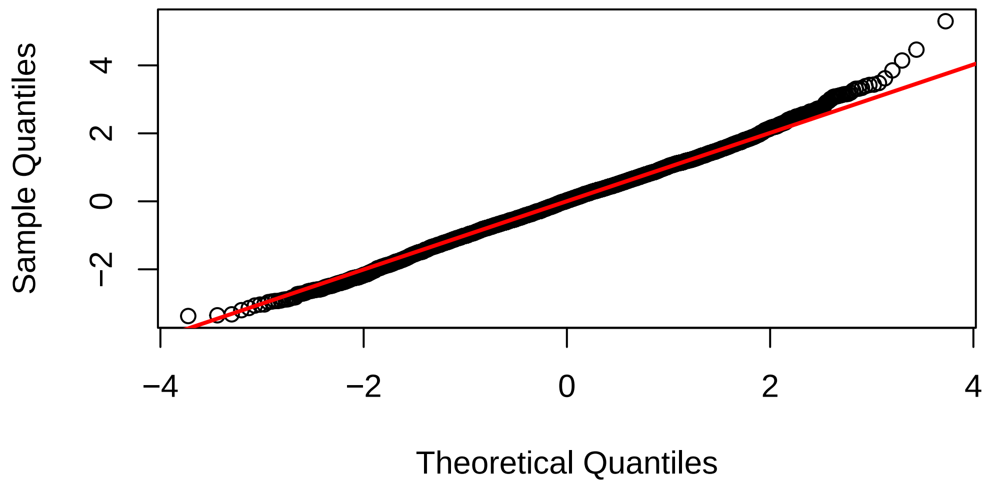

Figure 1: QQ-Plot of The Final Model

Figure 2 illustrates the relative risk with 95% confident interval between different variable groups and their reference group. Compared to individuals who married between ages 22-25 (reference group), those who married at 0-15 or above 30 may have a slightly higher number of children (relative risk), but due to their wide confidence intervals covering 1, the reliability of this estimate is low. In contrast, those who married between 15-18 and 18-20 are more likely to have more children than reference group, as their confidence intervals are relatively narrow and do not include 1, indicating a stronger association. For individuals who married between 20-22 or 25-30, the number of children appears to remain nearly unchanged compared to the reference group, but since their confidence intervals include 1, no definitive conclusion can be drawn. For literacy, compared to the reference group (literacy), the illiteracy group has more children, and the confidence interval is very narrow, implying that there is a high probability that the illiteracy group has more children than the literacy group.

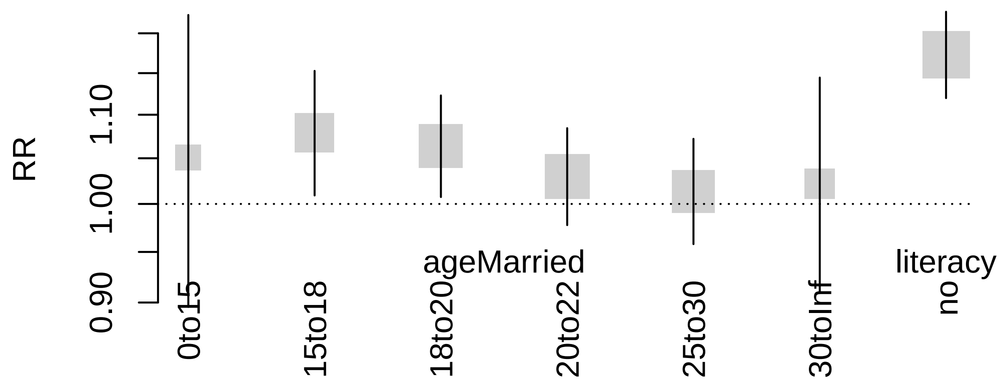

Figure 2: Relative Risk

More precise values can be seen in Table 2.

Table 2: Summary of The Final Model

	Estimate	Std. Error	z value	Pr(>|z|)	est	2.5 %	97.5 %
(Intercept)	-1.802	0.018	-99.076	0.000	0.165	0.159	0.171
ageMarried0to15	0.050	0.080	0.624	0.533	1.051	0.896	1.224
ageMarried15to18	0.076	0.034	2.236	0.025	1.079	1.009	1.153
ageMarried18to20	0.062	0.028	2.228	0.026	1.064	1.007	1.123
ageMarried20to22	0.029	0.026	1.105	0.269	1.030	0.978	1.084
ageMarried25to30	0.013	0.029	0.468	0.640	1.014	0.958	1.072
ageMarried30toInf	0.022	0.059	0.369	0.712	1.022	0.909	1.145
literacyno	0.159	0.024	6.770	0.000	1.173	1.120	1.228